

Model card doc, draft 2.1

Theory and Purpose:

Intrinsic value is an estimation of a stock's "real" price based on fundamental metrics. Since investor sentiment often skews the current price of a stock from its actual valuation, a benchmarking metric like Intrinsic value can be used to determine if an asset is overvalued or undervalued, leading to better investment decisions.

The intrinsic value nowadays is often calculated using Benjamin Graham's formula, shown in figure 1. However, this formula often uses a constant P/E ratio of 8.5 representing a company with no growth, and an additional constant of 4.4 representing the average yield of a AAA corporate bond. Thus, due to its use of constants it can be argued that this estimation alone can't be used to determine if a company is TRULY undervalued or overvalued.

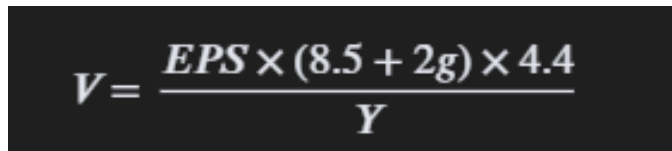

$$V = \frac{EPS \times (8.5 + 2g) \times 4.4}{Y}$$

Figure 1: Benjamin Graham's Intrinsic Value formula

Furthermore, the P/E ratio is also another indicator of a company's real value, since the higher a P/E ratio the less expected returns the asset will have. The use of these metrics also assumes that stocks perform the same, and exhibit the same values for such benchmarks across all sectors which could be false. Assets in slower moving sectors like utilities may have higher P/E ratios with Intrinsic Values being closer to current prices but that doesn't mean they aren't extremely overvalued or undervalued.

In conclusion, to determine the real Intrinsic value of an asset, metrics like P/E ratio and Intrinsic value difference must be compared with sector specific trends and historical patterns. This is when Machine Learning may play a vital role.

Model Architecture:

Translating a value-investing philosophy to Machine Learning is possible through a Stacked Machine Learning Pipeline, where the architecture is broken down into three distinct layers. Each layer accepts input, computes an output, and passes that output to the next layer which further computes using values. Below is a breakdown of the layers along with the various inputs and specific nuances.

1. Layer 1 Feature Engineering: calculates the "Graham" Intrinsic value of the stock, and accepts the P/E ratio from user input.
 - Inputs: price, EPS, P/E, expected growth, and AAA corporate bond Yield
 - Outputs: two primary features P/E Ratio and the IV% Difference vs price

2. Layer 2 Unsupervised Learning Model: using a Gaussian Mixture Model (GMM) to handle the "relativity" of different market sectors and valuations
 - Action: for each sector, the GMM identifies two natural clusters
 - Output: probabilities, telling the next layer how likely a stock is to belong to either the "Undervalued" cluster or the "overvalued" cluster
3. Layer 3 Supervised Learning Model: A Neural Network (Logistic Regression), trained on Historical data which compares P/E ratios, Graham Intrinsic Values, sector specific probabilities of clusters, and real outcomes of price movement between the years 2021–2025 which verifies when the GMM was actually right about its probability.
 - Training Label 1 = Winner: the stock grew over the past 4 years
 - Training Label 0 = Loser: the stock fell over the past 4 years
 - Action: the NN learns to "veto" the GMM results, if the GMM says a stock is 99% undervalued, but the NN sees a historical pattern of "Value Traps" in that sector.

The architecture then uses historical patterns to find which sectors have consistently produced cheaper stocks, and what a cheap stock in each sector even looks like. It uses the GMM to understand Sector Context and the NN to understand Historical Probability of Success.

Training Data Organization:

The data sources from Kaggle provide the following pieces of information between the years ranging from 2017 to 2021.

- Ticker symbol
- Name of the stock
- Name of the sector
- Price of the stock, accurate to last date when data was updated (2021)
- EPS, earnings per share, taken from 2021
- P/E ratio, taken from 2021
- Market Cap, taken from 2021
- Start value (price), during 2017
- End value (price), during 2021
- Median AAA Corporate bond yields during the year 2021

The following needs to be calculated based on the data provided from the Datasets.

- Growth rate over the last 5 - 6 years, or expected growth rate

The following is the only piece of data which requires the use of an API, possibly yfinance, to find the stock price movement between the years 2021 and 2025.

- Real valuation, "undervalued" if price went up between the date range and "overvalued" if it went down between the date range, assuming the market corrected the price

Static metrics like EPS and P/E, are taken from 2021, while the growth rate is based on a historical 4 years from 2017 - 2021. The idea is to simulate a real trading behavior during training, allowing the model to calculate Graham's IV, and use the GMM's probability to make an

assessment on the valuation of the stock, “overvalued” or “undervalued”. Then real historical data from 2021 to 2025 will be used to show how the market actually moved the stock price, showing the model if it was actually correct or not about the prediction, assuming the market corrected any price inconsistency. This will be the label on the training data.

Data Sources:

[s&p 500 EPS and PE ratio](#)

[Current Corporate AAA bond yield](#)

[s&p 500 growth rate price/returns](#)

Calculation References:

- ▶ [How to Calculate the Intrinsic Value of a Stock like Benjamin Graham! \(Step by Step\)](#)
- ▶ [How To Calculate Intrinsic Value \(Full Example\)](#)

- $\text{EPS (earning per share)} = \text{net income} / \text{number of shares}$
- $\text{P/E ratio} = \text{price per share} / \text{EPS}$
- $\text{Growth Rate} = (\text{end value} / \text{start value})^{1/4} - 1$
- Current yield of AAA bonds = find online on the yahoo moody tracker

Required User Inputs:

The values required to run the model are the following, these values are primarily used to calculate the Graham Intrinsic Value to price difference.

- Current price
- EPS (earning per share)
- P/E ratio
- Growth Rate
- Current yield of AAA bonds

Potential Model packaging:

```
def my_custom_formula(X):
    combined_feature = X[:, 0] * X[:, 1]
    return np.column_stack((X, combined_feature))

pipeline = Pipeline([
    ('custom_math', FunctionTransformer(my_custom_formula)),
    ('neural_net', Perceptron()) # Or any MLPClassifier
])

import joblib
joblib.dump(pipeline, 'full_model.joblib')
```